Mirror, Not Mask: Testing Whether Persona Self-Report Tracks Its Own Activation Geometry

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***Abstract*—Language models can be asked what persona they are currently adopting, and separately, their internal activations can be measured while they adopt it. Whether these two surfaces, what a model says about itself and what its representations actually look like, describe the same underlying structure or two loosely related ones is not obvious, and most prior work on persona representation builds one of these surfaces without testing its correspondence to the other. We address this directly using representational similarity analysis, a method borrowed from systems neuroscience, to compare the geometry of a language model's internal activations against two independently constructed self-report channels: free-text persona description and forced single-token self-rating. Using sixteen personas spanning five relational clusters and two deliberately confusable pair types, we extract activations at three pre-registered, depth-matched layers (approximately one quarter, one half, and three quarters of network depth) in three instruction-tuned models spanning two independently trained families and a parameter range from 1.5 billion to 7 billion: Qwen2.5-7B-Instruct, Qwen2.5-1.5B-Instruct, and Llama-3.2-3B-Instruct. In every model, activation geometry shows a strong, statistically significant correspondence with both self-report channels (self-description RSA up to 0.725, 0.672, and 0.817 respectively; self-rating RSA up to 0.779, 0.687, and 0.717) that survives partialing out the confound of prompt-text wording alone. Once corrected for each model's own split-half measurement reliability, the correspondence's magnitude is comparable across all three models, while the layer at which it peaks differs by architecture: both Qwen models peak at the deepest tested layer, Llama-3.2-3B peaks at the middle layer. This work is explicitly correlational and motivates, rather than performs, a causal follow-up test of whether this correspondence is load-bearing for model behavior.**

***Index Terms*—representational similarity analysis, mechanistic interpretability, persona representation, self-report, activation geometry, cross-model replication**

I. Introduction

Instruction-tuned language models are routinely asked to adopt a persona, a physician, a pirate, a customer service agent, and they comply fluently, producing text that a reader would recognize as being in character. At the same time, a growing body of mechanistic interpretability work has shown that this kind of identity or trait information leaves a measurable trace in a model's internal activations, recoverable as linear directions, clusters, or shared feature structure. These two observations, that models can describe their own persona in language and that their persona leaves a geometric signature in activation space, are usually reported separately, by different lines of work using different tools. What is rarely asked directly is whether they are the same structure viewed from two angles, or whether a model's self-report is closer to a performance layered on top of an internal state that may organize itself quite differently.

This question matters beyond curiosity about any single model. If a model's account of its own persona tracks the geometry of its activations closely, self-report becomes a cheap, interpretable proxy for an otherwise expensive internal measurement, and a plausible first signal in the broader project of model self-knowledge. If the two are weakly linked, then self-report is closer to a mask than a mirror: fluent, contextually appropriate, but representationally decorative, and any interpretability or alignment argument that leans on what a model says about itself needs to be made far more cautiously. Put plainly, the question this paper asks is whether a language model's self-described persona identity is a mirror of its internal representational state, or a mask placed over a state that need not resemble what is being said.

We test this directly rather than assume an answer, using representational similarity analysis (RSA), a method developed in systems neuroscience for exactly this kind of cross-modality comparison: relating the geometry of one measurement space (here, activations) to the geometry of another (here, self-report), without requiring the two spaces to share units, dimensionality, or even measurement modality. Two independently constructed self-report channels are compared against activation geometry: a judge-scored free-text persona description, and a judge-free forced single-digit self-rating read directly from output token probabilities. Both are compared against activation geometry extracted at three layers chosen in advance of seeing any result, placed at approximately one quarter, one half, and three quarters of each model's depth, so that the same relative developmental stages are compared across architectures of differing layer count philosophy. Throughout, prompt-text wording is treated as the single most important confound to rule out: an embedding-based text-only baseline is partialed out of every reported correlation, so that "personas use different words, and different words produce different activations" cannot by itself explain the result.

Rather than reporting this test in a single model, where any observed correspondence could plausibly be a property of one training run or one architecture rather than a general phenomenon, we replicate the identical pipeline, persona set, and pre-registered layer indices across three instruction-tuned

models spanning two independently trained families and a four and two thirds fold range in parameter count: Qwen2.5-7B-Instruct, Qwen2.5-1.5B-Instruct, and Llama-3.2-3B-Instruct. Every tokenizer- and chat-template-dependent mechanic (digit token identities, hook output shape, generation-point offset) is independently reverified for each model rather than assumed to transfer, a discipline that in practice caught a real layer-indexing bug before it entered any cross-model table and correctly diagnosed a genuine, model-specific weakness in one model's self-rating channel rather than silently accepting it.

Concretely, this paper contributes:

- A representational similarity test of persona self-report fidelity, using two independently constructed self-report channels and a text-wording confound control, replicated identically across three instruction-tuned language models from two model families.
- A statistically grounded primary finding: activation geometry corresponds significantly to both self-report channels in every model tested, at every pre-registered layer, surviving partialing out the prompt-text confound.
- A ceiling-normalized cross-model comparison showing that the correspondence's magnitude is comparable across a 1.5 billion to 7 billion parameter range once each model's own measurement reliability is accounted for, while the depth at which it peaks is architecture-dependent rather than fixed.
- An honest accounting of where the self-report channels diverge in reliability across models, used to motivate specific, falsifiable hypotheses for a planned causal follow-up study rather than to paper over the channel's known weaknesses.

The remainder of this paper reviews related work, describes the representational similarity methodology and the controls used to rule out the text-wording confound, reports the primary cross-model results, discusses their interpretation alongside genuine alternative explanations, and closes by scoping the claims this correlational study does and does not support.

II. Related Work

Interest in how language models represent identity, persona, and character has grown alongside the practical need to monitor and control these behaviors. Chen, Arditi, Sleight, Evans, and Lindsey [1] introduce *persona vectors*: activation-space directions extracted directly from natural-language trait descriptions, used to monitor and predict finetuning-induced personality drift and to screen training data before it is used. Their method constructs a geometric representation of a trait from a description of that trait, and demonstrates that the resulting direction is causally useful for steering and monitoring. It does not, however, ask whether a model's own self-report about which persona it is currently expressing corresponds to the geometry of its activations while expressing that persona; the construction runs in the opposite direction, from description to vector, rather than testing fidelity between a model's live self-report and its live internal state.

A second and closely related line of work examines how multiple personas coexist inside a single model. Danilov, Nourbakhsh, Marchenko Breneur, and Lamsiyah [2] use sparse autoencoders to show that role-play personas retain a shared, Assistant-associated feature core, while characters generated in a narrated story do not show the same shared core. This finding motivates a specific methodological choice in the present work: rather than selecting sixteen maximally distinct personas, our persona set deliberately includes pairs designed to be hard to separate, most directly a first-person persona and its narrated-story counterpart (for instance, *pirate* and *pirate_story*), precisely because Danilov et al. suggest these pairs sit close together in representation space for a principled reason rather than by accident. A representation that only separates obviously distinct personas would tell us comparatively little; one whose errors land specifically on these engineered hard pairs provides a positive control for fine-grained structure, not merely coarse categorical separation.

Separately from the persona-representation literature, representational similarity analysis itself originates in systems neuroscience, where Kriegeskorte, Mur, and Bandettini [3] introduced it as a way to relate the representational geometry of one measurement space, originally neural activity recorded across brain regions or species, to that of another space, such as behavioral judgments or computational model predictions, without requiring either space to share units or dimensionality with the other. Nili et al. [4] formalize the accompanying machinery used throughout the present study, including split-half noise-ceiling estimation as a way to separate a genuinely weak correspondence from a noisy measurement of a possibly strong one, a distinction that becomes central to the cross-model comparison reported in Section V. Within language-model interpretability specifically, layer-wise probing and representational-similarity studies by Tenney, Das, and Pavlick [5] and Voita, Sennrich, and Titov [6] established that progressively deeper transformer layers encode increasingly abstract linguistic structure, a finding this paper's layer-selection methodology builds on directly by sampling three fixed relative depths rather than exhaustively sweeping every layer.

Positioning the present work. Persona vectors are built from descriptions without testing self-report fidelity to the resulting geometry; the shared-persona-core finding motivates which personas belong in a stress test of fine-grained separability without itself performing a self-report correspondence test; and representational similarity analysis supplies exactly the right tool for the comparison neither of these prior studies makes. The present work sits at the intersection of these three threads: it pairs a representational-similarity test with identity and persona content specifically, uses two independently constructed self-report channels rather than one, controls explicitly for the confound of prompt-text wording through partial correlation, and replicates the full pipeline across two independently trained model families rather than reporting a single-model result. Every citation supporting this positioning was independently verified against its primary

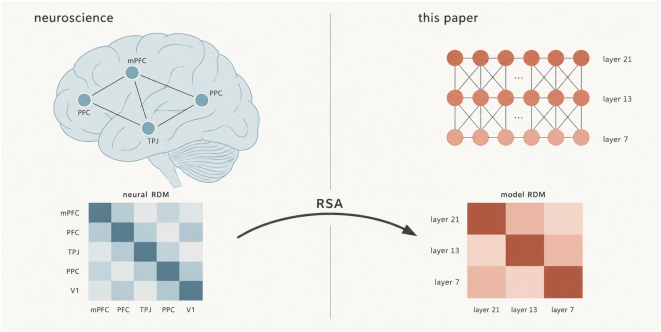


Fig. 1: Representational similarity analysis as a bridge between measurement spaces. Left: the method's origin in systems neuroscience, comparing a dissimilarity matrix built from one measured response space against a dissimilarity matrix built from a second, independent space. Right: the present study reuses the identical comparison, with activation geometry (RDM-A) standing in for the first measurement space and self-report (RDM-F, RDM-S) standing in for the second, bridged by the same Spearman-correlation-based RSA computation defined in Equation 1.

source before being included here, after an earlier, ChatGPT-authored literature summary used during project scoping was found to contain at least one fabricated attribution; references that could not be independently confirmed are not carried into this section.

III. Methods

A. Models and Depth-Matched Layer Selection

Three publicly available instruction-tuned decoder-only transformers are evaluated: Qwen2.5-7B-Instruct (hidden size 3584), Qwen2.5-1.5B-Instruct (hidden size 1536) [7], and Llama-3.2-3B-Instruct (hidden size 3072) [8], the last requiring explicit license acceptance on HuggingFace before weights could be loaded. All three models share an architectural detail that is convenient rather than arranged: each has exactly twenty-eight decoder layers, indexed 0 through 27. This allows a single set of absolute layer indices to refer to matched relative depths across all three models simultaneously, rather than requiring separately computed indices per architecture.

Layers were selected before any result was inspected, at approximately one quarter, one half, and three quarters of total network depth, corresponding to an early layer, a middle layer, and a late layer in each model's own processing. The exact indices used, hand-picked rather than derived from a rounding formula applied to depth fractions, are layer 7 (twenty-five percent depth), layer 13 (forty-six percent depth, the closest well-separated index to the midpoint), and layer 21 (seventy-five percent depth). These are reported as the `PRIMARY_LAYERS` throughout, and are held fixed across all three models so that any observed difference in where the correspondence is strongest reflects the model being studied rather than a shifting measurement point. Pre-registering these layers in advance, rather than selecting them after seeing which ones look best, is what keeps the depth-dependence finding reported in Section V an honest correlational observation rather than a post hoc, cherry-picked one.

TABLE I: Model specifications and shared layer indices

Model	Hidden size	Layers	Primary layers
Qwen2.5-7B-Instruct	3584	28	7, 13, 21
Qwen2.5-1.5B-Instruct	1536	28	7, 13, 21
Llama-3.2-3B-Instruct	3072	28	7, 13, 21

Before any extraction, four environment checks were independently rerun for each model rather than assumed to carry over from the previous run: the forward-hook output type (confirmed a plain tensor, not a tuple, in all three models), the identity of every digit token used by the self-rating channel (confirmed single token and mutually distinct across all nine digits in every tokenizer), the exact offset at which the formatted chat template reaches its generation point (confirmed to end with no trailing content before generation begins), and two hook-correctness checks: an alpha equals zero no-op check, confirming the extraction hook is a true no-op until deliberately activated, and a per-position delta-norm check, confirming that any targeted perturbation lands exactly at the intended sequence position and does not leak into neighboring positions. This full reverification was not a formality: it is what caught a real layer-indexing divergence in the 1.5 billion parameter run (Section III-E) before it entered any reported number.

B. Persona and Prompt Construction

Sixteen personas are used, held identical across all three models and their variants: *assistant*, *physician*, *professor*, *teacher*, *interrogator*, *critic*, *pirate*, *pirate_story*, *customer_service*, *drill_sergeant*, *poet*, *poet_story*, *therapist*, *comedian*, *librarian*, and *diplomat*. Personas are organized into five relational clusters, summarized in Table II, so that the resulting activation geometry can be compared against a meaningful prior structure rather than sixteen arbitrary, unrelated categories.

TABLE II: Persona taxonomy by relational cluster

Cluster	Personas
Anchor (1)	assistant
Expert (3)	physician, professor, librarian
Warm (3)	customer_service, teacher, therapist
Cold (4)	drill_sergeant, interrogator, diplomat, critic
Playful (5)	pirate, poet, pirate_story, poet_story, comedian

Pairwise relationships among the sixteen personas are additionally classified into four engineered pair types: close (five pairs drawn from the same cluster), far (three pairs drawn from maximally distinct clusters), speaker-mode (two pairs contrasting a first-person persona against its narrated-story counterpart, following the shared-persona-core motivation from Section II), and surprise (four pairs deliberately constructed to be difficult to separate on the basis of surface wording

alone, used as a positive control for fine-grained activation structure in Section V-A). Each persona is instantiated with two independently written system-prompt variants, which is what enables both a leave-one-variant-out separability check and a split-half reliability estimate: without a second variant, there would be no way to distinguish a model's genuine representational reliability from a single noisy sample. Eight elicitation prompts, held identical across personas, models, and variants, are used to generate the free-text descriptions and collect self-ratings.

C. Self-Report Channels and the Representational Dissimilarity Matrix Framework

Four independently constructed representational dissimilarity matrices (RDMs) are built per model, each a sixteen by sixteen matrix summarizing how dissimilar every pair of personas is under one measurement channel:

- **RDM-A**, activation geometry: the measurement being explained, built from last-token hidden states captured at each of the three primary layers during a forward pass over the elicitation prompts.
- **RDM-P**, a prompt-text embedding baseline: the confound being explicitly controlled for, built from an embedding model applied directly to the system-prompt text with no activations involved.
- **RDM-F**, free-text self-description: personas are elicited to describe themselves in open text, and the resulting descriptions are scored by an independent judge model rather than the model under study, avoiding the circularity of a model grading its own persona-following.
- **RDM-S**, forced single-digit self-rating: a judge-free channel built directly from output token probabilities, in which the model rates how well a candidate persona description matches a target persona on a nine-point scale, and the expected value under the resulting digit-token distribution is read off directly rather than relying on any generated text or external judge. Diagonal self-ratings (a persona rated against its own description) are included as a built-in validity check, since a well-calibrated channel should rate these close to the maximum of nine.

This four-channel design is what allows the primary result to be corroborated twice: RDM-F and RDM-S are independently constructed, using different elicitation formats and different scoring mechanisms (an external judge model versus the subject model's own logits), so agreement between them is evidence that neither channel's result is an artifact of its particular construction.

D. Representational Similarity Analysis and Statistical Controls

For a pair of RDMs X and Y, representational similarity is computed as the Spearman rank correlation between their vectorized upper triangles,

$$RSA(X, Y) = \rho_{\text{Spearman}}\big(\text{vec}(X),\, \text{vec}(Y)\big). \tag{1}$$

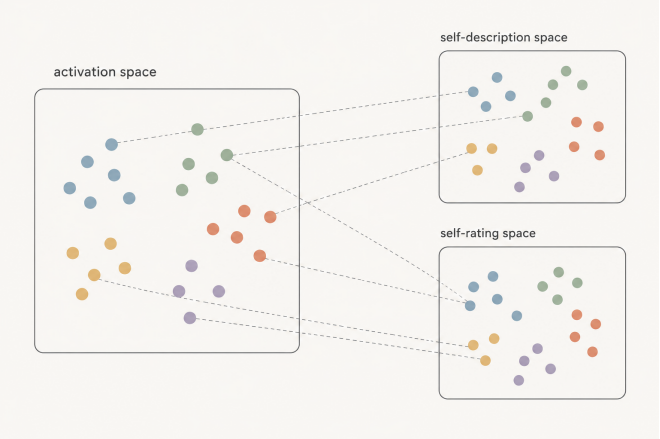


Fig. 2: Activation space and the two self-report spaces, each a sixteen-point cloud colored by the same five relational clusters from Table II. Dashed lines connect a representative subset of matching personas across spaces. RSA measures whether the overall arrangement of points, not any single coordinate, agrees across activation space and each self-report space.

Spearman correlation is used rather than Pearson correlation because it makes no assumption about the linear scale of dissimilarity in either space, which is appropriate given that activation-space distance and judge-scored text dissimilarity are not expected to share a common linear unit.

Four controls accompany every reported RSA value, each addressing a specific, nameable alternative explanation rather than serving as rigor for its own sake.

Permutation test. Every reported RSA value is tested against a null distribution built by repeatedly permuting the persona labels of one RDM before recomputing RSA, establishing significance against a proper null rather than merely observing that RSA exceeds zero.

Shuffled-label control. As a separate sanity check beyond the permutation test, RSA is recomputed once more under a single, fixed label shuffle; a value indistinguishable from zero confirms that the RSA computation itself is not trivially inflated by an implementation artifact independent of genuine structure.

Partial RSA. The single most important control in this pipeline. Because different personas are necessarily described with different words, and different words produce different activations through the embedding layer alone, a raw correspondence between activation geometry and self-report could in principle be explained entirely by shared sensitivity to surface wording rather than to persona identity itself. Partial RSA removes this possibility by controlling for RDM-P, the text-only embedding baseline, using the standard partial correlation formula

$$\rho_{A,F|P} = \frac{\rho_{A,F} - \rho_{A,P}\,\rho_{F,P}}{\sqrt{\left(1-\rho_{A,P}^2\right)\left(1-\rho_{F,P}^2\right)}}, \tag{2}$$

with the analogous expression used for $\rho_{A,S|P}$. Every primary

result reported in Section V is given both raw and with this control applied.

Split-half reliability and the noise ceiling. RDM-A is built separately from each of the two system-prompt variants, and the RSA between these two variant-specific activation RDMs is taken as an estimate of the measurement's own internal reliability, its noise ceiling C at a given layer. This makes it possible to distinguish a genuinely weak correspondence from a noisy measurement of a possibly strong one, and is the basis for the ceiling-normalized cross-model comparison in Section V-D, computed as

$$\widehat{RSA}(L) = \frac{RSA(L)}{C(L)}. \tag{3}$$

E. Per-Model Verification Protocol

Rather than running the pipeline once and copying its logic forward, every model-specific mechanic listed in Section III-A was independently reverified for each of the three models. This caught two real issues worth reporting directly, since both bear on how the Section V results should be read. In the 1.5 billion parameter Qwen run, an automated layer-selection formula computing indices as the nearest integer to each depth fraction produced layer 14 rather than the hand-picked layer 13 used in the 7 billion parameter run, a silent divergence caught by auditing printed figures against the hand-picked indices before any number entered a cross-model table, and corrected by hardcoding the shared index set and fully re-extracting activations. In the Llama-3.2-3B run, direct probing of the model's top-predicted output tokens at the generation point confirmed that this model's chat template ends its generation point with two newline characters rather than the single trailing space used by the Qwen template, and that this structural difference, not a measurement error, is responsible for a markedly lower confidence rate in that model's self-rating channel specifically; this is reported honestly as a model-specific limitation in Section V rather than patched or silently excluded.

IV. A Note on Interpretation

Before turning to results, it is worth stating plainly how they should be read. The inspection framework used throughout this study is correlational by design. Every reported correspondence is derived from a single trained checkpoint per model, a fixed persona set, and a fixed pair of self-report elicitation formats; none of the analyses reported below involve any activation-level intervention. Recurring patterns are deliberately sought across two independent self-report channels and three independently trained models to increase confidence in the interpretations offered in Section VI, but establishing that this correspondence is causally load-bearing for model behavior, rather than merely co-occurring with it, will require intervention-based techniques such as activation patching, which are explicitly out of scope for the present study and are reserved for a planned follow-up.

V. Results

Before turning to statistical tests, Figures 3–5 show, for each model, all four representational dissimilarity matrices at that model's peak layer side by side. The visual resemblance between RDM-A and the two self-report matrices, RDM-F and RDM-S, relative to their shared resemblance to the text-only baseline RDM-P, is the qualitative picture the statistics in the remainder of this section make precise.

A. Persona Expression and Activation Separability

Judge-scored persona expression (zero to three scale) and leave-one-variant-out activation separability at layer 13 are summarized in Table III. The judge's pass-rate threshold of 0.85 was met only by Llama-3.2-3B-Instruct, whose role-persona pass rate of 0.85 also corresponds to the cleanest separation between the *assistant* baseline (mean 0.38) and the role personas of any of the three models. Manual inspection of raw generations in every experiment confirmed in-character text even for personas the judge under-scored, consistent with a known judge-calibration limitation that under-rewards quieter, professionally registered personas (*customer_service*, *librarian*, *diplomat*, *interrogator*) rather than a genuine failure of persona expression.

TABLE III: Persona expression pass rate and activation separability

Model	S2 pass rate[a]	S5 accuracy[b]	Length-expr. r
Qwen2.5-7B	0.66	0.906	0.43
Qwen2.5-1.5B	0.57	0.742	0.15
Llama-3.2-3B	0.85	0.914	-0.11

[a]All role personas, threshold ≥ 0.85. [b]Layer 13, chance $= 0.0625$ for $K = 16$.

Separability at layer 13 is well above chance in every model (chance accuracy 0.0625 for sixteen personas) but the structure of the remaining confusion errors differs sharply by model. In Qwen2.5-7B, confusions land on three of the four engineered surprise pairs (*drill_sergeant* confused with *interrogator* in five of eight held-out trials, *physician* with *therapist* in five of eight, *librarian* with *therapist* in two of eight). In Llama-3.2-3B, confusions land precisely on the speaker-mode surprise pairs (*pirate_story* with *pirate* in five of eight, *poet_story* with *poet* in four of eight), the same clean pattern as the larger Qwen model. In Qwen2.5-1.5B, by contrast, confusions are diffuse rather than landing on the designed pairs: *therapist* acts as a generic attractor, pulling in misclassifications from four otherwise unrelated personas (*physician*, *teacher*, *customer_service*, *librarian*), indicating coarser sub-cluster structure at this parameter scale rather than a scaled-down version of either other model's confusion pattern.

B. Baseline Representational Similarity and Reliability

The embedding-based text baseline, RSA between activation geometry and prompt text alone with no self-report involved, is significant at every layer in every model (Table IV), confirming

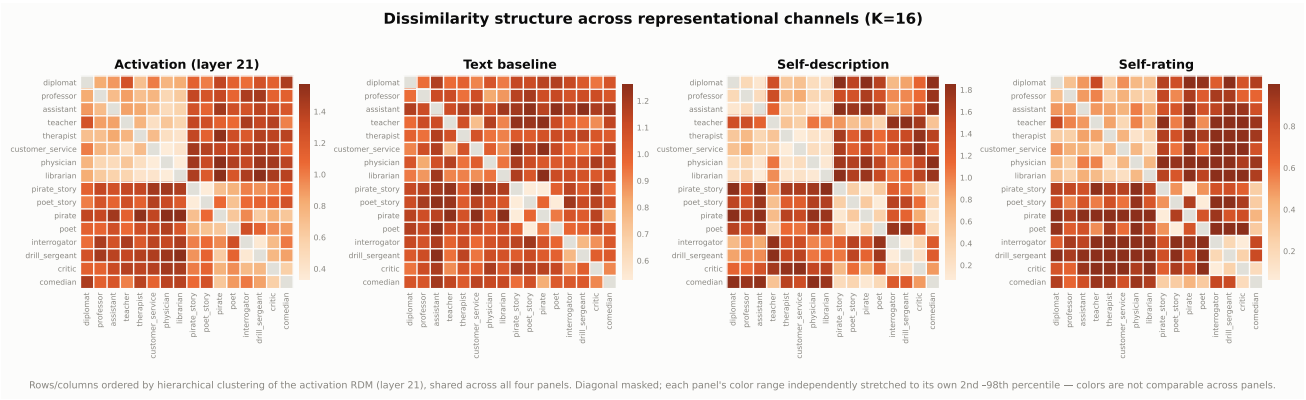


Fig. 3: Qwen2.5-7B-Instruct: all four representational dissimilarity matrices at layer 21, the model's peak layer. Darker cells indicate greater dissimilarity between the row and column personas; persona order (fixed by hierarchical clustering of the activation RDM) is identical across all four panels. Each panel's color range is independently stretched to its own distribution, so colors are not comparable across panels, only the block structure within each is.

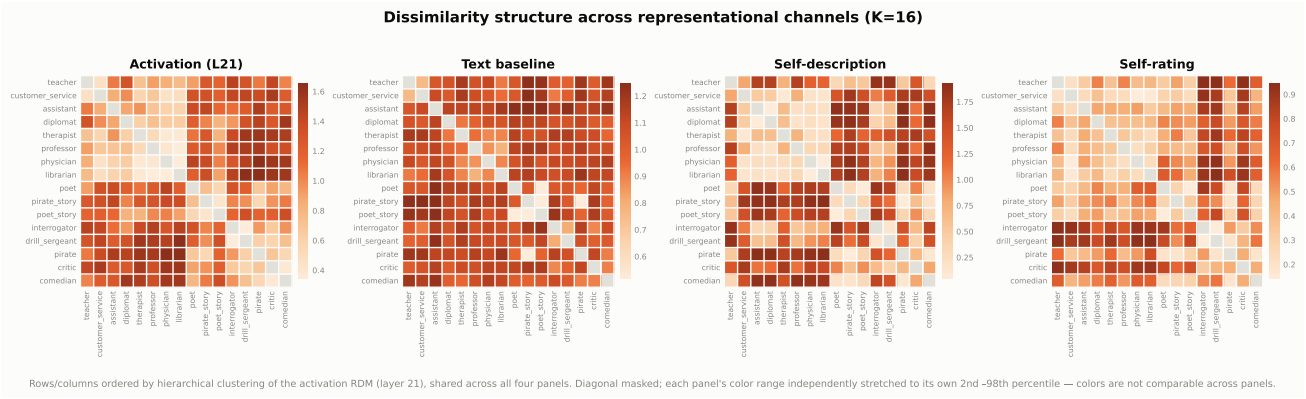


Fig. 4: Qwen2.5-1.5B-Instruct: all four representational dissimilarity matrices at layer 21, the model's peak layer. Note the visibly coarser block structure relative to Figure 3, consistent with the diffuse separability confusions reported in Section V-A.

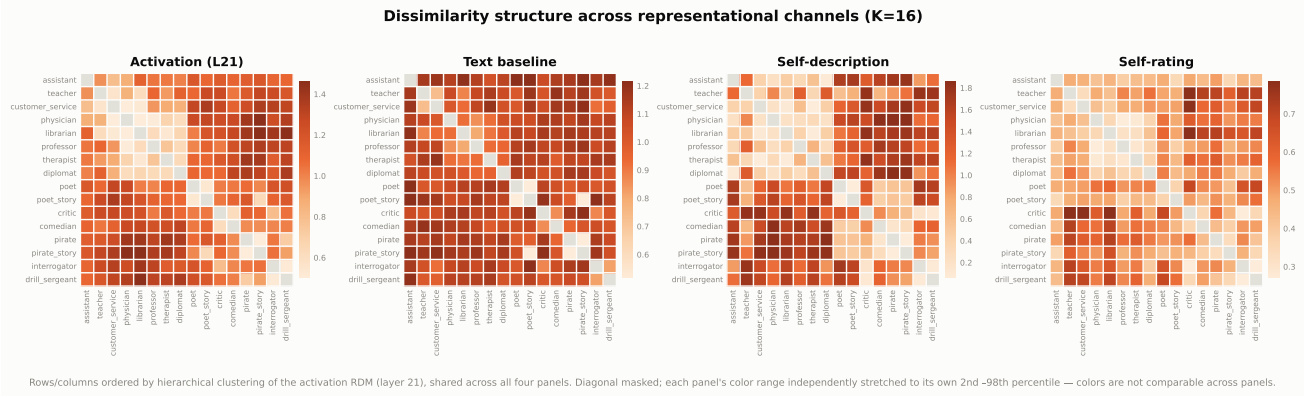


Fig. 5: Llama-3.2-3B-Instruct: all four representational dissimilarity matrices at layer 13, the model's peak layer (note this is the middle, not the deepest, layer, unlike both Qwen models).

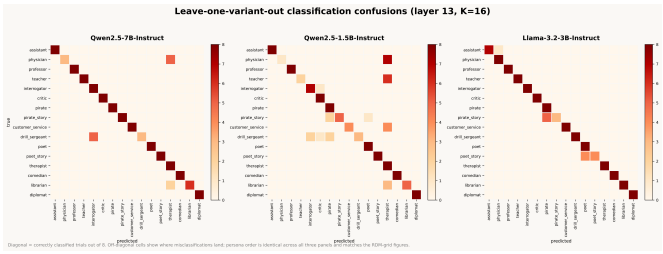


Fig. 6: Leave-one-variant-out confusion structure at layer 13 for all three models. Confusions in Qwen2.5-7B and Llama-3.2-3B concentrate on the engineered surprise pairs; confusions in Qwen2.5-1.5B are diffuse, with *therapist* acting as a generic attractor.

that prompt-text wording is a genuine confound requiring explicit control rather than a theoretical concern. The shuffled-label control RSA is close to zero at every layer in every model (Qwen2.5-7B: -0.047; Qwen2.5-1.5B: -0.090; Llama-3.2-3B: -0.063, all at layer 13), confirming the RSA computation itself is not inflated by an implementation artifact.

TABLE IV: Text-baseline RSA and split-half noise ceiling by layer

Model		Layer 7	Layer 13	Layer 21
Qwen2.5-7B	Baseline RSA	0.622	0.463	0.404
	Ceiling	0.769	0.837	0.967
Qwen2.5-1.5B	Baseline RSA	0.537	0.499	0.397
	Ceiling	0.529	0.669	0.857
Llama-3.2-3B	Baseline RSA	0.557	0.419	0.427
	Ceiling	0.913	0.948	0.968

The split-half noise ceiling differs substantially across models, and this difference is itself informative. Llama-3.2-3B's activation geometry is far more internally consistent across the two independently written prompt variants than either Qwen model, especially at the shallowest layer (ceiling 0.913, against 0.769 for Qwen2.5-7B and 0.529 for Qwen2.5-1.5B at layer 7), a difference entirely independent of the self-report question and relevant to interpreting raw RSA magnitude fairly in Section V-D. Qwen2.5-1.5B additionally shows the one non-strictly-increasing baseline pattern is absent here but its ceiling is markedly lower at every layer than either larger model, consistent with a smaller model's activation geometry being intrinsically less split-half reliable, independent of anything about persona or self-report.

C. Primary Result: Activation-Self-Report Correspondence

The paper's headline finding is reported in Table V: raw and partial RSA, with prompt-text wording partialed out, between activation geometry and both self-report channels, at all three layers, in all three models. Every reported value is significant under the permutation test ($p \leq 0.017$ in every case, and $p \leq 0.0008$ in all but one cell).

Two patterns are visible directly in Table V. First, in every model and every layer, the correspondence survives partialing out prompt-text wording, falling by a modest, consistent margin rather than collapsing toward zero; this is the direct evidence against the alternative explanation that "personas use different words, and different words produce different activations" is sufficient to explain the raw correspondence. Second, the two Qwen models peak at the deepest tested layer, 21, with strictly increasing RSA across depth, while Llama-3.2-3B peaks at the middle layer, 13, and is slightly lower at the deepest layer than at the middle one for both self-report channels. This is not noise: the direction of the effect, middle-layer peak versus deep-layer peak, is consistent across both independently constructed self-report channels within Llama-3.2-3B, and is addressed directly in Section VI-B.

Agreement between the two independently constructed self-report channels, measured as RSA between RDM-F and RDM-S directly, is 0.723 in Qwen2.5-7B, 0.456 in Qwen2.5-1.5B, and 0.667 in Llama-3.2-3B. The weaker agreement in the 1.5 billion parameter model is traceable to that model's self-rating expected-value calibration (mean 7.09 against a target of 9, minimum 3.54) being notably worse than either larger model's (Qwen2.5-7B: mean 8.87, minimum 6.97; Llama-3.2-3B: mean 7.16, minimum 6.15), rather than to a problem with the free-text description channel specifically.

D. Ceiling-Normalized Cross-Model Comparison

Raw RSA magnitude alone makes the smaller Qwen2.5-1.5B model appear uniformly weaker than Qwen2.5-7B at every layer. Given the substantially lower split-half ceiling reported in Table IV for that model, this comparison is not fair on its own terms: a lower ceiling means the measurement itself is noisier, independent of whether the underlying correspondence is weaker. Table VI reports the ceiling-normalized score from Equation 3, dividing each raw RSA value by that model's own split-half reliability at the same layer.

Once corrected for reliability, the ordering by raw parameter count disappears. The smallest model tested, Qwen2.5-1.5B, shows the single highest ceiling-normalized self-description score in the entire table (0.925 at layer 13), exceeding both larger models at that layer. Llama-3.2-3B's self-description correspondence remains strong at every layer after correction (0.723, 0.862, 0.811), landing between the two Qwen models at most layers rather than standing out as an outlier of either family. Self-rating's ceiling-normalized correspondence is weaker relative to self-description specifically at the deepest layer for Llama-3.2-3B (0.718, the lowest value among the nine self-rating cells involving that model), which is consistent with the self-rating confidence weakness documented in Section III-E rather than a broader degradation of the underlying representational correspondence.

VI. Discussion

A. Generality Across Scale and Family

The correspondence between activation geometry and self-report is not a property of one model's particular training run. It replicates across a four and two thirds fold reduction in parameter count within the same model family (Qwen2.5-7B to Qwen2.5-1.5B) and, independently, across a change of

TABLE V: Primary result: activation-self-report RSA, raw and with text-wording partialed out

Model		Self-description (RDM-F)			Self-rating (RDM-S)		
		Layer 7	Layer 13	Layer 21	Layer 7	Layer 13	Layer 21
Qwen2.5-7B	Raw	0.464	0.616	**0.725**	0.426	0.650	**0.779**
	Partial	0.341	0.549	**0.683**	0.279	0.586	**0.743**
Qwen2.5-1.5B	Raw	0.417	0.619	**0.672**	0.237	0.476	**0.687**
	Partial	0.262	0.529	**0.610**	0.102	0.400	**0.652**
Llama-3.2-3B	Raw	0.660	**0.817**	0.785	0.440	**0.717**	0.695
	Partial	0.563	**0.777**	0.737	0.334	**0.678**	0.652

All values significant under permutation test, $p \leq 0.017$; peak layer per row in bold.

TABLE VI: Ceiling-normalized RSA, all three models, both channels

	Self-description			Self-rating		
	Layer 7	Layer 13	Layer 21	Layer 7	Layer 13	Layer 21
Qwen2.5-7B	0.603	0.736	0.750	0.554	0.777	0.806
Qwen2.5-1.5B	0.788	**0.925**	0.784	0.448	0.711	0.802
Llama-3.2-3B	0.723	**0.862**	0.811	0.482	**0.756**	0.718

model family entirely (Qwen to Llama), using the identical pipeline, persona set, and pre-registered layer indices in both comparisons. The ceiling-normalized comparison in Table VI is the more defensible basis for comparing magnitude across these models: once corrected, smaller or differently trained models do not show a systematically weaker correspondence, they show a noisier, lower-ceiling measurement of a comparably strong one. This is a meaningfully different and more cautious claim than "bigger models have stronger self-knowledge," and it is the claim the data actually support.

B. Architecture-Dependent Locus of the Correspondence

The one finding visible only because three models were tested, not two, is that the correspondence does not peak at the same relative depth across architectures. Both Qwen models peak at the deepest tested layer; Llama-3.2-3B peaks at the middle layer and is lower at the deepest layer, consistently across both independently constructed self-report channels. Three layers is not a full depth sweep, so the true peak for any of these models, Llama-3.2-3B in particular, is located only to the nearest of three sampled points rather than pinpointed. Still, the direction and consistency of the Llama result across two independent channels is too large to be dismissed as a random fluctuation of the permutation test, and it directly motivates a specific, falsifiable claim for future causal work: the layer at which persona representation is causally load-bearing, if it is load-bearing at all, should not be assumed to transfer from one architecture to another on the basis of a single model's result.

C. Channel Robustness Asymmetry

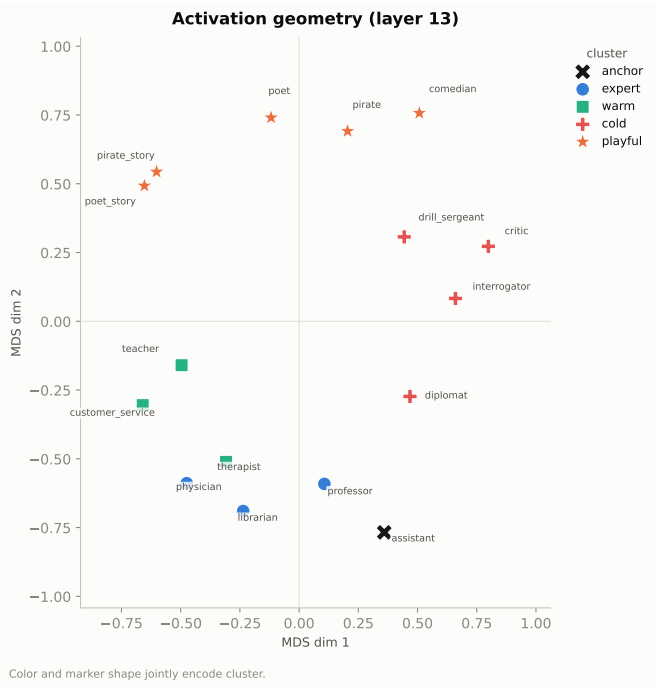
Across all three models, self-description is the more dependable channel: well-behaved raw magnitude, strong agreement with the independently constructed self-rating channel in two of three models, and stable ceiling-normalized performance. Self-rating is significant everywhere but is more sensitive to model-specific instruction-following calibration, most visibly in Llama-3.2-3B, where direct probing of the model's own top-predicted tokens confirmed that a structural difference in that model's chat-template generation point, not a measurement error, degrades confidence in the self-rating channel specifically. A plausible account is that free text sits closer to a model's natural output distribution, while a terse, forced single-digit answer is comparatively more sensitive to how directly each model's own instruction-tuning calibrates confident, compliant single-token responses under its native chat-template formatting.

D. Alternative Explanations

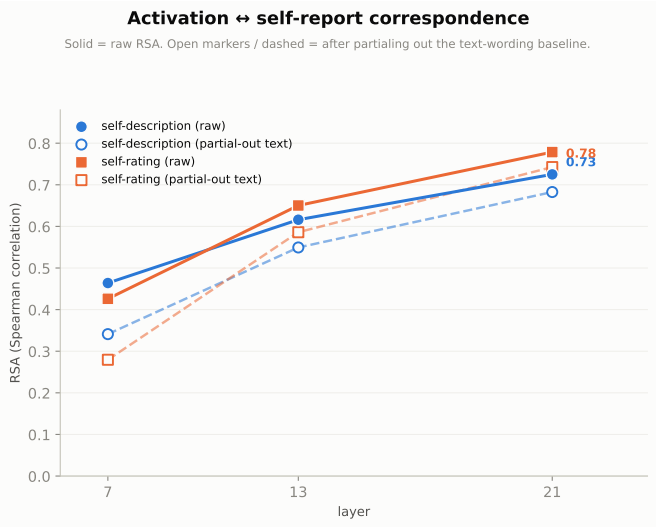
The interpretation offered above, that self-report and activation geometry share real, persona-specific structure, is not the only account consistent with the reported numbers, and it is worth pausing on genuine alternatives rather than treating the primary result as self-evidently settled.

One alternative concerns the partial-RSA control itself. Partialing out RDM-P removes the component of correspondence explained by a linear embedding-space representation of prompt-text wording, but an embedding model is itself an imperfect proxy for the full space of surface-level cues a transformer's own tokenizer and attention mechanism could exploit. It remains possible that some residual component of the partial-RSA correspondence reported in Table V reflects a surface-level regularity the embedding baseline does not fully capture, rather than a persona-specific representational structure as such. This cannot be ruled out by the present design and would require a baseline built directly from the subject model's own token-level representations of the prompt, rather than an external embedding model, to address more directly.

A second, related alternative concerns the ceiling-normalization procedure in Section V-D. Dividing by a model's own split-half reliability corrects for measurement noise, but it could in principle also reward a model whose activation geometry is reliable for reasons unrelated to persona

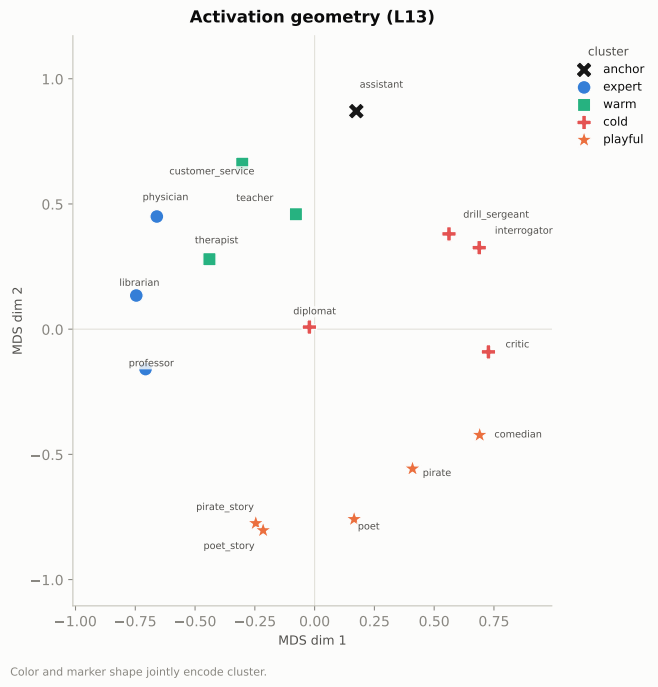


(a) Activation geometry (2-D MDS)

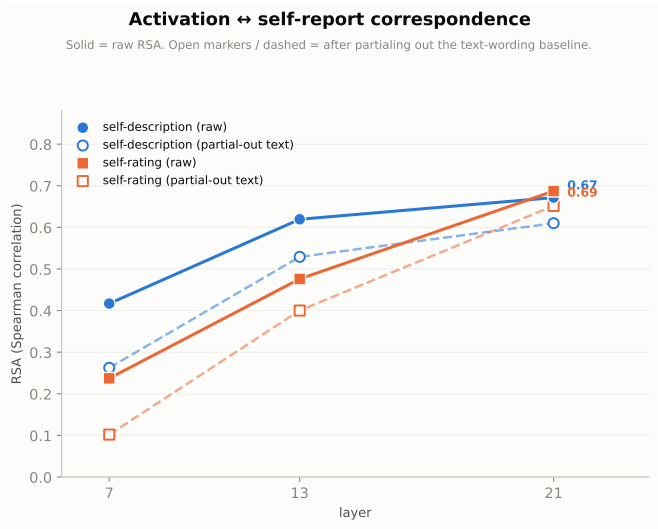


(b) Correspondence by layer

Fig. 7: Qwen2.5-7B-Instruct: activation geometry, colored by the five relational clusters of Table II, alongside the resulting layer-wise correspondence with both self-report channels, raw and with text wording partialed out.

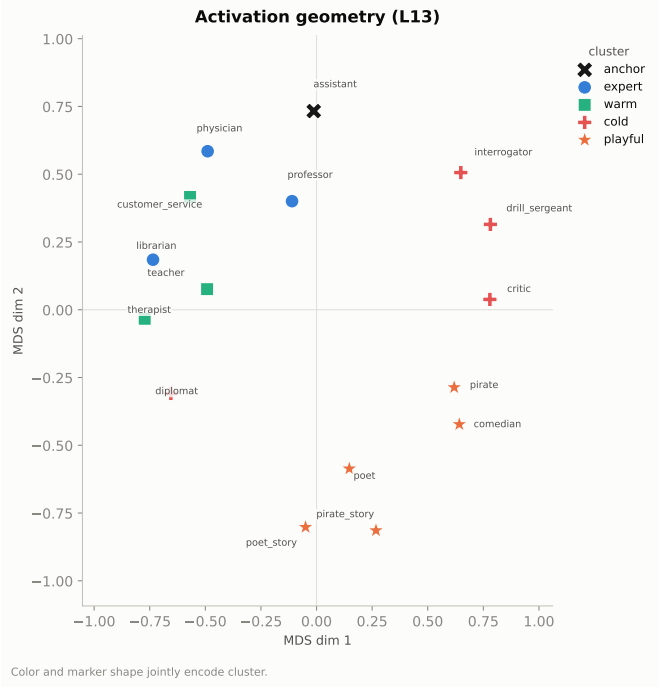


(a) Activation geometry (2-D MDS)

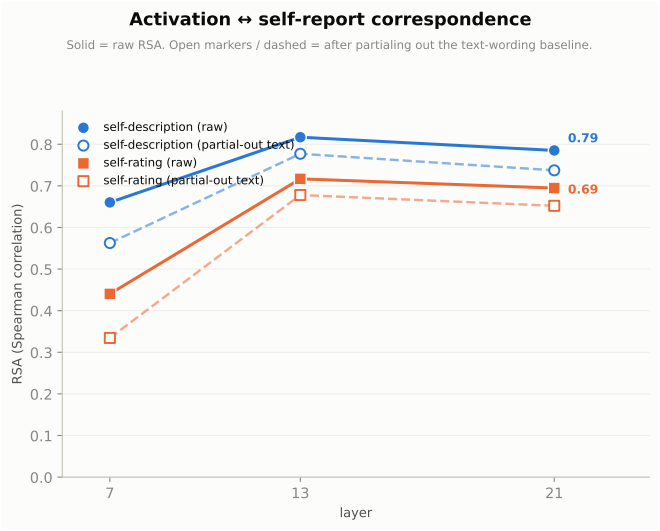


(b) Correspondence by layer

Fig. 8: Qwen2.5-1.5B-Instruct: activation geometry and the resulting layer-wise correspondence with both self-report channels. Cluster boundaries in (a) are visibly less separated than in Figure 7a, consistent with the lower S5 accuracy reported in Table III.



(a) Activation geometry (2-D MDS)



(b) Correspondence by layer

Fig. 9: Llama-3.2-3B-Instruct: activation geometry and the resulting layer-wise correspondence with both self-report channels. Panel (b) shows the middle-layer peak directly, in contrast to both Qwen models' deepest-layer peak visible in Figures 7b and 8b.

content, for instance, a geometry dominated by a small number of large, broadly stable coordinates that are reliable across prompt variants without being specifically informative about persona identity. Under this reading, Llama-3.2-3B's notably higher ceiling at every layer (Table IV) might partly reflect this kind of stable-but-generic structure rather than an unusually persona-faithful representation, which would mechanically inflate its ceiling-normalized score in Table VI without the underlying self-report correspondence actually being stronger in a meaningful sense. Directly inspecting the dimensionality and stability of the dominant activation directions at the layers in question would help adjudicate between these two readings.

A third alternative concerns the middle-layer peak observed in Llama-3.2-3B. Rather than reflecting a genuine architecture-dependent difference in where persona representation is strongest, as proposed in Section VI-B, the pattern is also consistent with that model's chat template injecting a fixed, persona-independent date preamble into every system prompt (Section III-E), which could in principle interact differently with representations at different depths than Qwen's template, which contains no such preamble. This was assessed as low risk during data collection because the preamble is identical across every persona and condition and should not itself carry persona-discriminating information, but it was not independently proven to have zero effect on the layer-wise shape of the correspondence, and the possibility that it contributes to the specific layer at which Llama-3.2-3B peaks cannot be fully excluded by the present design.

None of these alternatives make the reported correspondence any less real as a statistical fact, each value in Table V remains significant under a proper permutation null regardless of which interpretation is correct, but they do relocate the open question from whether an effect exists to what, specifically, it reflects, which is exactly the kind of question a causal follow-up using targeted intervention could adjudicate directly.

E. Scope and Limitations

The claims in this paper should be read within the scope of what was actually tested, not inflated beyond it. All analyses were performed on a single trained checkpoint per model, a fixed persona set of sixteen, two system-prompt variants per persona, and three pre-registered layers rather than a full depth sweep. No claim is made or supported about a causal role for this correspondence in model behavior; this is explicitly a correlational study, scoped deliberately to motivate rather than substitute for the causal, intervention-based follow-up discussed throughout Section VI. Robustness to substantially different phrasings, elicitation formats, or a larger and differently constructed persona taxonomy is untested. Three models, spanning two independently trained families, constitutes strong initial evidence of generality rather than an exhaustive demonstration; a fully general claim that this correspondence holds for instruction-tuned transformer language models as a class would require a broader and more architecturally diverse model sample than the present study provides.

Two limitations are specific to individual models rather than shared across all three. In Qwen2.5-1.5B, self-rating expected-value calibration is notably worse than in either larger model, and leave-one-variant-out separability confusions are diffuse rather than landing on the designed surprise pairs, both read as genuine signal-quality degradation at this parameter scale rather than hidden or averaged away. In Llama-3.2-3B, sixty-six percent of self-ratings fell below the pre-registered confidence threshold, traced through direct top-token probing to a genuine, model-specific weakness in terse single-digit instruction-following under that model's own chat-template structure; self-rating RSA for this model should accordingly be read with more caution than its self-description RSA, or than either Qwen model's self-rating channel.

F. Implications for Cross-Family Generality

What this study buys, concretely, is not a definitive answer to whether self-report and activation geometry are causally linked, but a correlational claim that was pre-registered at the layer level, independently re-verified at the extraction-mechanics level for each model rather than assumed to transfer, controlled for the most obvious confound of prompt-text wording, checked against a proper statistical null, corroborated by two independently constructed self-report channels, and replicated across both model scale and model family. The broader question this raises, whether a correspondence between what a model says about itself and what its internals actually look like is a general organizing property of instruction-tuned transformer language models or a coincidence of how a small number of models happen to have been trained, cannot be settled by three models alone. What can be said is that the correspondence did not weaken or disappear when the test was moved outside the model family where it was first observed, which is the single strongest form of evidence a purely correlational, single-paper study of this kind can offer.

VII. Concluding Remarks

This paper asked whether a language model's self-described persona identity is a mirror of its internal representational state or a mask placed over one. The answer, within the scope of the three models and sixteen personas tested, is closer to a mirror than a mask: activation geometry corresponds significantly to both an independent judge-scored description channel and a judge-free self-rating channel, in every model, at every pre-registered layer, after partialing out the confound of prompt-text wording. This correspondence is comparable in magnitude once corrected for each model's own measurement reliability, even though it peaks at different relative depths in different architectures, a genuinely new finding visible only because three models were compared rather than one or two.

What this paper does not establish is whether this correspondence is causally load-bearing for model behavior, or whether it would survive a model deliberately trained to decouple the two. Both questions are reserved for a planned causal follow-up using targeted activation intervention at the specific layers identified here, a study this paper is intended to motivate rather than preempt. If that work confirms what the present correlational evidence suggests, the mirror reading will have earned a stronger claim than correlation alone can offer; if it does not, the alternative explanations raised in Section VI-D are exactly where that investigation should begin.

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